# Glaucoma Detection by Combining SLO and OCT Fundus Images: A Pretrained Two-Branch EfficientNet

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November 2026

## 1 Introduction

**Task.** Glaucoma is a chronic optic-nerve disease and a leading cause of irreversible blindness, so automated screening from retinal imaging is valuable. We address one problem from the FairDomain benchmark [1]: *given two grayscale images of the same eye, an SLO fundus image and an OCT-derived fundus image, predict whether the eye has glaucoma* (binary classification). A solution outputs a probability $P(\text{glaucoma})$ for each image pair. It is judged by the area under the ROC curve (AUC) on a fixed held-out test set and, because the data carry demographic attributes, by how evenly that performance is spread over gender, race and ethnicity groups (Section 4.3).

### 1.1 The data

**Contents.** The release holds 5,000 samples in a fixed split of 3,000 training and 2,000 test samples. Each sample is one `.npz` file with the two images (`slo_fundus` and `oct_fundus`; each $224 \times 224$, 8-bit, single channel), the label `glaucoma` (0 or 1), and patient attributes: age, gender, race, ethnicity, language, marital status and the visual-field mean deviation (MD). A CSV file indexes the split and the attributes. No segmentation masks are included, so the task is classification only.

**What the images look like.** Figure 1 shows random samples of each class. Both modalities show the optic disc and the retinal vessels around it. The SLO fundus image is darker, with uneven illumination and dark patches; the OCT fundus image is brighter and smoother. The two modalities therefore differ clearly in appearance, but *the two classes do not look different to the eye*, and simple intensity statistics do not separate them either: over 300 random training images the mean pixel value is 73 (no glaucoma) against 70 (glaucoma) for SLO, and 169 against 169 for OCT. The signal is subtle structural information that a model has to learn.

**Labels and demographics.** Glaucoma is the positive class in 44% of both splits (1,317 of 3,000 training and 887 of 2,000 test samples), a mild imbalance that is one reason we report AUC rather than accuracy. The demographic groups are strongly imbalanced (Table 1): 73% of samples are female, 77% white, 16% Black, 8% Asian and 4% Hispanic. Only 83 test images belong to Hispanic patients, so any estimate for that group is very noisy.

**Clinical variables.** MD summarises a visual-field test (in dB; 0 is normal, more negative means more field loss). It averages $0.1 \pm 0.8$ in non-glaucoma and $-8.2 \pm 5.7$ in glaucoma training samples, so it almost encodes the label: giving it to the model raises test AUC to 0.958 (Appendix F). We treat it as a leakage risk and use it only in one clearly labelled upper-bound run, never in a reported

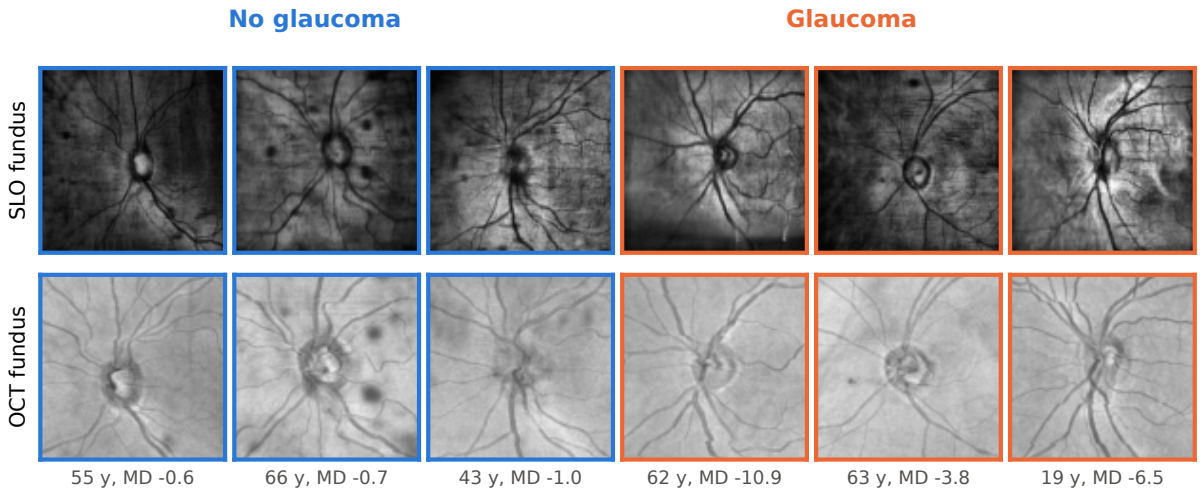



**Figure 1:** Randomly chosen training samples (seed 0, not hand-picked) for each class. Top row: SLO fundus; bottom row: OCT fundus. Under each pair: the patient's age and visual-field mean deviation (MD, dB). Blue frame: no glaucoma; orange frame: glaucoma.

| Split | $N$ | Glaucoma | Female | White | Black | Asian | Hispanic |
|---|---|---|---|---|---|---|---|
| Training | 3000 | 1317 (44%) | 2204 (73%) | 2308 (77%) | 467 (16%) | 225 (8%) | 121 (4%) |
| Testing | 2000 | 887 (44%) | 1449 (72%) | 1522 (76%) | 301 (15%) | 177 (9%) | 83 (4%) |

**Table 1:** Dataset composition. Percentages are of each split.

comparison. Age differs by class (57.8 against 64.4 years on average) but adding it did not help (Appendix F).

**What an engineer should keep in mind.**

- **Small data:** after holding out validation data only 2,550 training images remain, so overfitting is the main difficulty (Appendix C).
- **Small test set:** 2,000 images give AUC confidence intervals of about ±0.02; differences smaller than that between two models are not meaningful.
- **No official validation set:** we hold out a stratified 15% of the training images for all model and configuration selection (Section 4.1).
- **Two different views:** a model trained on one modality transfers poorly to the other (Appendix F), so the modalities must be handled separately and combined.
- **Fairness-aware evaluation:** results should be reported per demographic group, not only overall.

**Our approach and result.** Our final model is a *two-branch EfficientNet-B0*: one ImageNet-pretrained branch per modality, features concatenated, trained for 15 epochs with a plain recipe. It reaches a test AUC of 0.827 (95% CI 0.808–0.844) against 0.717 for a from-scratch ResNet18 baseline. It was the best of 13 configurations on test AUC, so the figure is somewhat optimistic: other two-branch runs of the same architecture score 0.777–0.819 (Section 4.5, Appendix C). The main text

describes only this model and the experiments that justify it; the other experiments (a 13-model backbone comparison, a training-recipe study with ablations, fusion variants, a foundation model, fairness losses, cross-domain transfer, small ensembles) are documented with their motivation and setup in the appendices.

**Contributions.** (1) A simple and cheap pipeline (about 5 minutes per run) for SLO+OCT glaucoma classification, with bootstrap confidence intervals and per-group results. (2) A controlled demonstration, on the same backbone and recipe, that using both modalities beats either alone (0.827 against 0.758 and 0.764). (3) A training-recipe study in which stronger augmentation hurt, including a hypothesis of ours that the ablations refuted (appendices). (4) A measure of how noisy single-run results are on this data (re-runs and seeds), which tells how much to trust small differences.

## 2 Related Work

- **Datasets from the Harvard Ophthalmology AI Lab.**
  - Harvard Glaucoma Fairness [2]: glaucoma data with 2D and 3D images. It proposes Fair Identity Normalization (FIN), which makes feature importance equal across groups.
  - Harvard Eye Fairness [3]: 30,000 subjects with 3D OCT. It proposes Fair Identity Scaling (FIS), which weights the loss by individual and by group.
  - FairCLIP [4]: fairness for vision–language models on related data.
  - FairDomain [1] (our data): fairness under domain shift, for segmentation and classification. It treats image types as domains and proposes a fair identity attention (FIA) module.
- **Fairness metrics.** This line of work commonly reports group-wise AUC, worst-group AUC and equity-scaled AUC (ES-AUC).
- **Recent glaucoma work (not reproduced by us).**
  - FairEnc [5]: fair vision–language model on SLO images with clinical notes.
  - Fair-Eye Net [6]: combines fundus photos, OCT-derived metrics and visual-field indices, with a fairness constraint.
  - RED-Sphere [7]: domain generalisation across populations for fundus disease.
- **Models and tools we use.**
  - Final model: EfficientNet [9] with ImageNet [19] weights from `timm` [20]. Training: AdamW [22] with a one-cycle schedule [23].
  - Also tested (Appendix B): ResNet [8], DenseNet [10], RegNet [11], ConvNeXt [12], ViT [13], DeiT [14], Swin [15], CoAtNet [16], MaxViT [17], DINOv2 [18].
  - Appendices only: label smoothing [27], drop-path [26], mixup [24], CutMix [25], gradient reversal [28].

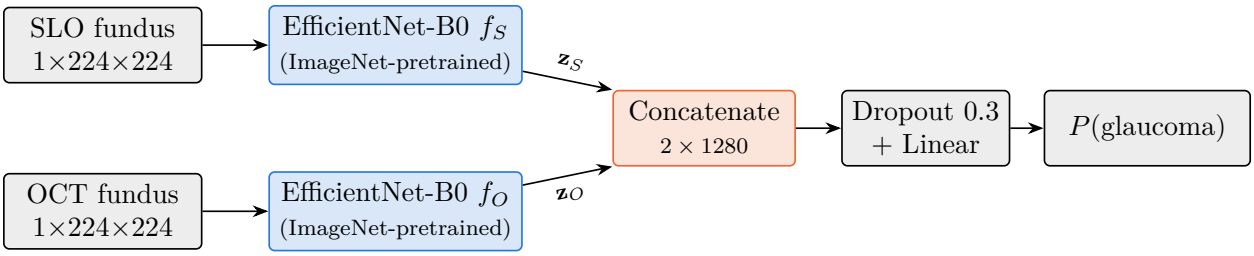



**Figure 2:** The final model: a two-branch late-fusion network with an EfficientNet-B0 backbone per modality.

# 3 Method

## 3.1 Problem

- Input: a pair $(\mathbf{x}_S, \mathbf{x}_O)$ of $224 \times 224$ single-channel images (SLO fundus, OCT fundus).
- Output: $p(y = 1 \mid \mathbf{x}_S, \mathbf{x}_O)$, the probability of glaucoma ($y \in \{0, 1\}$).
- Demographic attributes $g$ are used only to measure fairness.

## 3.2 Two-branch EfficientNet-B0

- One EfficientNet-B0 backbone per modality ($f_S$, $f_O$). Separate weights, initialised from ImageNet-1k, 4.0 M parameters each (Figure 2).
- The pooled features (1280 values per branch) are concatenated, then dropout (0.3), then one linear layer with two outputs.
- `timm` adapts the pretrained first layer to one input channel.
- Pixels are scaled to $[0, 1]$, then normalised as $(x - 0.5)/0.25$.
- Other fusion designs (early, gated, cross-attention) did not clearly help (Appendix D).

## 3.3 Training recipe ("plain recipe")

- Loss: cross-entropy.
- Optimiser: AdamW with a one-cycle schedule (15% warm-up). Learning rate $5 \times 10^{-4}$, weight decay $10^{-2}$, batch size 32, 15 epochs, gradient clipping 1.0.
- Augmentation: random horizontal flip and mild brightness/contrast jitter. Nothing else.
- No EMA, no label smoothing, no drop-path.
- The epoch with the best validation AUC is kept (epoch 13 of 15).
- Stronger recipes (more augmentation, EMA, label smoothing, drop-path, mixup) were tried later and did not beat this one reliably (Appendix C).

## 3.4 Test-time augmentation

- Optional: average the prediction on an image with the prediction on its horizontal flip.
- We report it in a separate column. The headline number does *not* use it.

| Hyper-parameter | Value |
|---|---|
| Backbone | EfficientNet-B0 $\times 2$ (ImageNet-1k, `ra_in1k` weights) |
| Optimiser / schedule | AdamW, one-cycle (15% warm-up), gradient clip 1.0 |
| Learning rate, weight decay | $5 \times 10^{-4}$, $10^{-2}$ |
| Batch size, epochs | 32, 15 |
| Head dropout | 0.3 |
| Augmentation | horizontal flip, brightness/contrast jitter (scale $\pm 10\%$, shift $\pm 0.025$) |
| Epoch selection | best validation AUC |

**Table 2:** Hyper-parameters of the final model.

# 4 Experiments

## 4.1 Data splits

- The data are described in Section 1.1 (Table 1).
- Validation set: a stratified 15% of the 3,000 training images (450). It is used to choose the epoch.
- The model trains on the other 2,550 images.
- The 2,000 test images are never used for training or for automatic selection.

## 4.2 Implementation and hyper-parameters

- PyTorch [21] with `timm` [20]; fp32; one Tesla P100 (16 GB). One training run takes about 5 minutes.
- Hyper-parameters are those of Section 3.3 (Table 2).

## 4.3 Evaluation protocol

- Main metric: AUC on the 2,000 test images. It needs no threshold and is not affected by the 44/56 class balance.
- Uncertainty: bootstrap 95% confidence interval over test images (1,000 resamples). Intervals are about $\pm 0.02$ wide. Differences smaller than that are not meaningful.
- Fairness metrics:
  - AUC inside each gender, race and ethnicity group.
  - Worst-group AUC: the lowest AUC over gender and race groups with at least 100 test samples.
  - Equity-scaled AUC (ES-AUC), computed per attribute:

$$\text{ES-AUC} = \frac{\text{AUC}}{1 + \sum_g |\text{AUC} - \text{AUC}_g|}. \tag{1}$$

- ES-AUC is our own implementation of the equity-scaled AUC used in the Harvard fairness benchmarks. Absolute values may differ from the original definition.

| Model | Input | Test AUC | 95% CI | + flip TTA | ES-AUC race | Worst-group AUC |
|---|---|---|---|---|---|---|
| ResNet18, two-branch, from scratch (baseline) | SLO+OCT | 0.717 | [0.694, 0.739] | 0.725 | 0.696 | 0.706 |
| EfficientNet-B0, SLO only | SLO | 0.758 | [0.736, 0.778] | 0.762 | 0.727 | 0.723 |
| EfficientNet-B0, OCT only | OCT | 0.764 | [0.742, 0.784] | 0.771 | 0.750 | 0.753 |
| **Final: EfficientNet-B0, two-branch** | SLO+OCT | 0.827 | [0.808, 0.844] | 0.828 | 0.815 | 0.816 |
| Same model, re-run with similar settings | SLO+OCT | 0.811 | [0.790, 0.830] | 0.814 | 0.792 | 0.804 |
| Same architecture, stronger recipe + drop-path | SLO+OCT | 0.817 | [0.797, 0.835] | 0.820 | 0.793 | 0.803 |
| Same architecture, stronger recipe, flips only | SLO+OCT | 0.819 | [0.799, 0.836] | 0.824 | 0.785 | 0.799 |
| Same architecture, stronger recipe, 3 seeds (mean ± s.d.) | SLO+OCT | 0.798 ± 0.003 | – | – | – | – |

**Table 3:** Main results on the 2,000 test images. The final model is in bold. "Stronger recipe" is described in Appendix C. TTA = horizontal-flip test-time averaging.

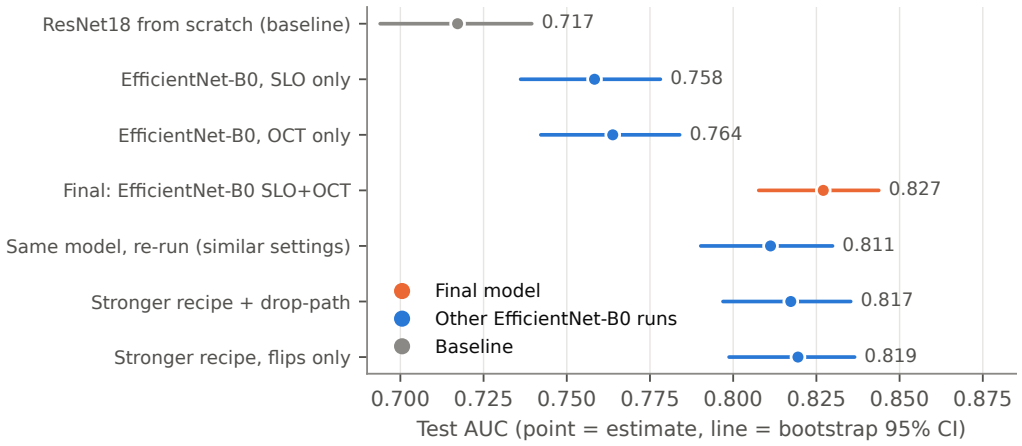



**Figure 3:** Test AUC with bootstrap 95% confidence intervals for the baseline, the single-modality models, the final model and other runs of the same architecture.

### 4.4 Baselines

- **ResNet18 from scratch:** same two-branch design, random initialisation, plain recipe.
- **Single-modality EfficientNet-B0:** same backbone and plain recipe, SLO only or OCT only.
- **Other runs of the same architecture:** a re-run with similar settings, and runs with stronger recipes (Appendix C). They show how much the result varies.

### 4.5 Results

Table 3 and Figure 3 show the main results.

- **Both modalities matter.**
  - Same backbone and recipe: SLO only 0.758, OCT only 0.764, both 0.827.
  - The gain (+0.06 to +0.07) is well outside the confidence intervals.
  - The two images carry different information: a model trained on one modality transfers poorly to the other (Appendix F).
- **Pretrained EfficientNet versus the baseline.**

| $n$ (test) | Female 1449 | Male 551 | Asian 177 | Black 301 | White 1522 | Non-Hisp. 1917 | Hispanic 83 |
|---|---|---|---|---|---|---|---|
| ResNet18 from scratch (baseline) | 0.721 | 0.706 | 0.711 | 0.735 | 0.710 | 0.723 | 0.556 |
| EfficientNet-B0, SLO only | 0.742 | 0.798 | 0.757 | 0.723 | 0.764 | 0.761 | 0.680 |
| EfficientNet-B0, OCT only | 0.759 | 0.777 | 0.765 | 0.770 | 0.753 | 0.766 | 0.704 |
| Final: EfficientNet-B0, two-branch | 0.824 | 0.835 | 0.823 | 0.816 | 0.827 | 0.830 | 0.740 |

**Table 4:** Test AUC within demographic groups.

  - From-scratch ResNet18: 0.717. Final model: 0.827 (95% CI 0.808–0.844). TTA adds 0.001.
  - This gap mixes pretraining and architecture. Every pretrained backbone we tried beat the scratch baseline (Appendix B), but we did not train scratch versions of the other backbones. So we cannot credit pretraining alone.

- **How stable is 0.827?**
  - The final model was the best of 13 configurations on test AUC. Swin-Tiny had the best validation AUC. So 0.827 is somewhat optimistic.
  - A re-run with similar settings scores 0.811. Stronger recipes score 0.798 (3-seed mean), 0.817 (drop-path) and 0.819 (flips only).
  - All our other two-branch EfficientNet-B0 runs score between 0.777 and 0.819. The gap to 0.827 is inside the confidence intervals.
  - We read the typical level of this architecture as about 0.80–0.82.

- **Fairness across groups** (Table 4).
  - Final model: AUC 0.816–0.835 across gender and race groups. Worst-group AUC 0.816 (baseline: 0.706). ES-AUC for race: 0.815.
  - The single-modality models are lower in every group.
  - The Hispanic group (83 test images) has the lowest AUC (0.740) but a very wide confidence interval. We make no claim about it.
  - Fairness-specific losses did not help (Appendix E).

### 4.6 Limitations

- The test set has 2,000 images. Confidence intervals are about ±0.02.
- The final model was picked because it had the highest test AUC among 13 configurations. Its figure is optimistic. The re-run (0.811) suggests by how much.
- The final model is one run with one seed.
- Automatic selection used validation data only, but we read test results while planning later experiments.
- Appendix C found that flips alone (no rotation, scale or shift) with the stronger recipe score about 0.02 higher than that recipe with geometric augmentation. We did not combine the helpful changes.

- ES-AUC is our own implementation.
- We did not implement the published FIN, FIS and FIA methods and did not compare with published numbers.

# 5 Conclusions

- **Result.** A two-branch EfficientNet-B0 (one ImageNet-pretrained branch per modality, concatenated features) reaches 0.827 test AUC (95% CI 0.808–0.844). The from-scratch ResNet18 baseline reaches 0.717.
- **What mattered.**
  - Using both modalities: about +0.06 to +0.07 AUC over either alone.
  - Pretrained backbones: every one we tried beat the scratch baseline.
  - Performance is similar across gender and race groups.
- **Strengths.** Simple and cheap: standard pretrained backbone, concatenation fusion, about 5 minutes per run.
- **Weaknesses.**
  - The 0.827 figure is probably optimistic. Other two-branch runs of the same architecture score 0.777–0.819.
  - The small test set cannot separate the top models.
  - Stronger augmentation hurt. The helpful changes we found were not combined.
- **Future work.**
  - Combine the helpful recipe changes and test them with several seeds.
  - Implement FIN, FIS and FIA faithfully and compare on the full dataset.
  - Fine-tune retinal foundation models.
  - Use the two image types as source and target domains with domain adaptation, since transfer between them is poor.
  - Use cross-validation over the full data to reduce noise from a single test split.

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# A Overview of all experiments

- Besides the final model we ran four batches of experiments: 51 training runs plus three small ensembles.
- All runs use the same data, the same split and the same validation protocol (Section 4.1). The epoch is chosen on validation AUC. Nothing was selected automatically on test AUC.
- Table 5 lists the batches. The next appendices give the details.

| Appendix | Question (motivation) | Setup | Main outcome |
|---|---|---|---|
| B Backbones | Which architecture family and modality choice work best? | 13 configurations, plain recipe, 15 epochs | Pretrained 0.747–0.827 vs scratch 0.717; EfficientNet-B0 best (0.827); both modalities help |
| C Training recipe | How to fight overfitting on 2,550 images? | Stronger recipe on Swin-Tiny and EfficientNet-B0; mixup, drop-path, LR, schedule; 4 ablations | Drop-path helps (0.817 vs 0.798); strong geometric augmentation hurts (flip-only 0.819) |
| D Fusion / foundation | Is smarter fusion or a self-supervised ViT better? | Gated fusion, cross-attention, DINOv2-S, Swin-scale recipe | No gain: 0.812, 0.795, 0.809 vs $0.808 \pm 0.004$ |
| E Fairness losses | Can fairness-aware training raise worst-group performance? | Group reweighting, FIS-like loss, adversarial race head | No consistent gain (0.787, 0.806, 0.805) |
| F Domain shift, tabular | How different are the two modalities? Do clinical variables help? | Train on one modality, test on the other; add age / MD | Transfer 0.68–0.70; age no gain; MD 0.958 (not comparable) |
| G Small ensembles, more data | Does averaging 3 models, or more training data, help? | Three 3-model ensembles; train+val and 5-fold runs | Ensembles 0.816–0.823 (about +0.02 over members); more data no gain |

**Table 5:** The experiments behind and around the final model.

# B Backbone and modality comparison (sweep 1)

- **Motivation.** Find out how much pretraining, the architecture family (CNN, transformer, hybrid) and the use of both modalities matter.
- **Setup.**
  - 13 configurations, *plain* recipe: cross-entropy, AdamW, one-cycle, 15 epochs, flips and mild brightness jitter. Learning rates: Table 6.
  - Late fusion (one backbone per modality), unless marked "early" (2-channel input to one backbone). Single-modality runs use one branch.

  - Epoch chosen by validation AUC.

- **Results** (Table 11, Figure 4).
  - Every pretrained model beat the from-scratch ResNet18 (0.717): EfficientNet-B0 0.827, Swin-Tiny 0.807, DenseNet121 0.803, CoAtNet-0 0.793, ViT-Small 0.790, MaxViT-Tiny 0.778, DeiT-Small 0.768, ResNet50 0.768 (early fusion 0.747), ConvNeXt-Tiny 0.767.
  - Models between 0.79 and 0.83 have almost fully overlapping confidence intervals. No architecture family is clearly best.
  - Modality (EfficientNet-B0, Figure 5): SLO only 0.758, OCT only 0.764, both 0.827.
  - ResNet50: late fusion 0.768, early fusion 0.747 (inside the intervals).
  - Swin-Tiny had the best *validation* AUC (0.844) but not the best test AUC. The 450-image validation set is noisy, so the recipe study (Appendix C) uses several seeds.

- **Takeaway.** Pretrained two-modality models work. The architecture family is secondary. We chose EfficientNet-B0: best test result and cheapest strong model (about 5 minutes per run). Choosing it on test AUC makes its figure optimistic.

| Backbone | Peak LR | Batch size | Weight decay |
|---|---|---|---|
| ResNet18 (scratch) | 3e-4 | 64 | 1e-2 |
| ResNet50 | 2e-4 | 32 | 1e-2 |
| EfficientNet-B0 | 5e-4 | 32 | 1e-2 |
| DenseNet121 | 2e-4 | 32 | 1e-2 |
| ConvNeXt-Tiny | 1e-4 | 32 | 5e-2 |
| ViT-S / DeiT-S / Swin-T | 5e-5 | 32 | 5e-2 |
| CoAtNet-0 | 1e-4 | 32 | 5e-2 |
| MaxViT-Tiny | 5e-5 | 16 | 5e-2 |

**Table 6:** Hyper-parameters of the first sweep.

# C Training recipe study and ablations (sweeps 2–3)

- **Motivation.** All models overfit. The plain EfficientNet-B0 reaches a training loss of about 0.008 while validation AUC stops improving early (Figure 6). We tried to regularise harder.

- **Setup.**
  - *Stronger recipe*: on-GPU rotation $\pm 15°$, scale $\pm 10\%$, shift $\leq 5\%$; label smoothing 0.1; EMA of the weights (decay 0.995); $5\times$ learning rate on the new head; 25 epochs.
  - First on Swin-Tiny (best validation AUC in sweep 1), 3 seeds.
  - Then on EfficientNet-B0 with variants: 12 epochs; mixup/CutMix; drop-path 0.2 with head dropout 0.5; lower learning rate ($2\times10^{-4}$); gated fusion. Plus four other backbones: EfficientNet-B2, B3, RegNetY-016, DenseNet121.
  - Rule for the configuration used in the train+val and 5-fold runs (validation AUC only): a variant replaces the plain stronger recipe only if its validation AUC beats the 3-seed mean by more than max(0.005, seed s.d.). Drop-path won: 0.852 against a mean of 0.832, margin 0.013.

- **Results** (Table 12 has the full list).

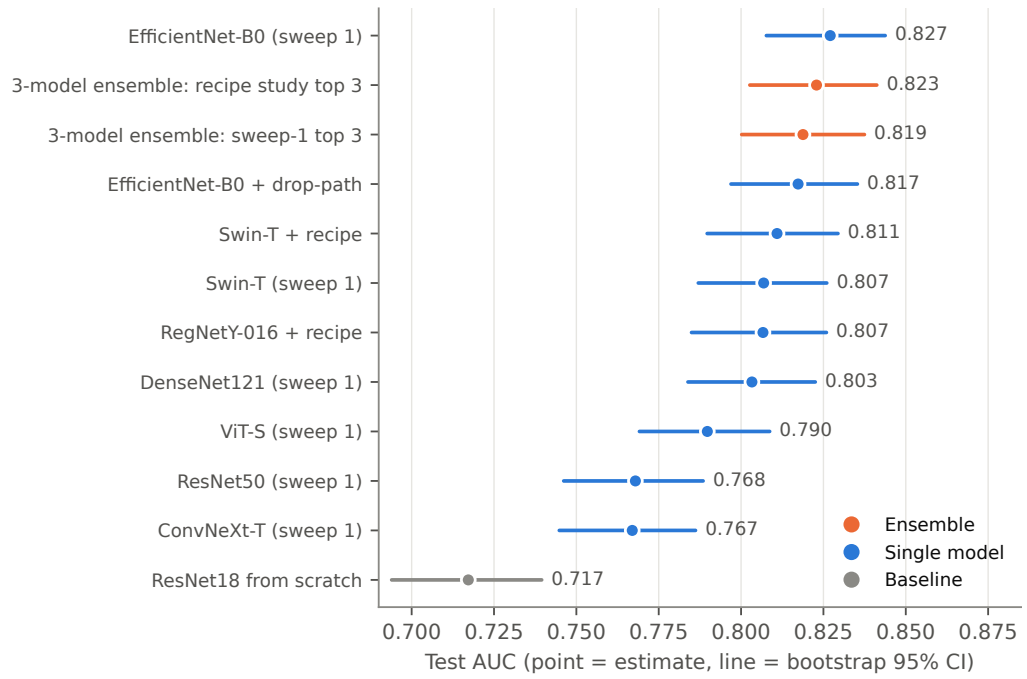



**Figure 4:** Test AUC with bootstrap 95% confidence intervals for the main single models and two 3-model ensembles.

  - Swin-Tiny: unchanged. $0.808 \pm 0.004$ over 3 seeds, against 0.807 in sweep 1.
  - EfficientNet-B0: *dropped* from 0.827 to $0.798 \pm 0.003$.
  - Drop-path with dropout helped: 0.817.
  - No help: mixup/CutMix 0.800, lower learning rate 0.790, 12-epoch schedule 0.777.
  - Other backbones were no better than EfficientNet-B0: B2 0.796, B3 0.779, RegNetY-016 0.807, DenseNet121 0.781.

- **Ablation: why did the recipe hurt EfficientNet?** (Table 7; one seed each)
  - Our first guess: the EMA code copies batch-norm statistics from the live model instead of averaging them. That could hurt batch-norm CNNs but not the layer-norm Swin.
  - EMA off: AUC fell to 0.786. The guess was *refuted*; EMA helps.
  - Flips and mild jitter only (no rotation, scale, shift): 0.819, up from 0.798. This is about 7 seed standard deviations. Strong geometric augmentation is the main culprit.
  - Head learning rate 1× instead of 5×: 0.807, a smaller gain.
  - All three choices reverted (approximate sweep-1 setup): 0.811, below the original 0.827. So part of that 0.827 is probably single-seed luck.
  - Possible reason for the augmentation result (not tested): the images may be spatially registered, so rotations and shifts could remove position-dependent cues.
  - Not run: the combination of the helpful changes (flip-only augmentation, drop-path, head learning rate 1×). It is the most promising next experiment.

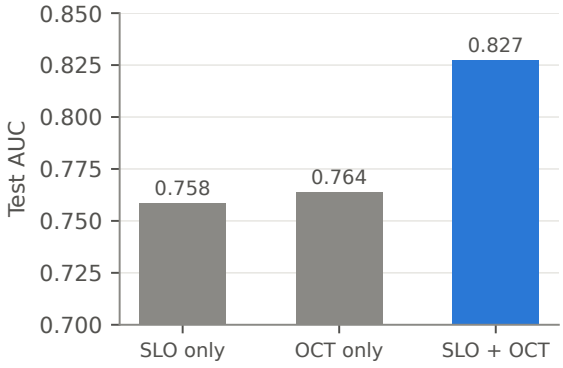



**Figure 5:** Contribution of each modality (EfficientNet-B0, plain recipe).

- **Takeaway.** Most standard regularisers did nothing. Drop-path was the one clear win. Geometric augmentation was harmful.

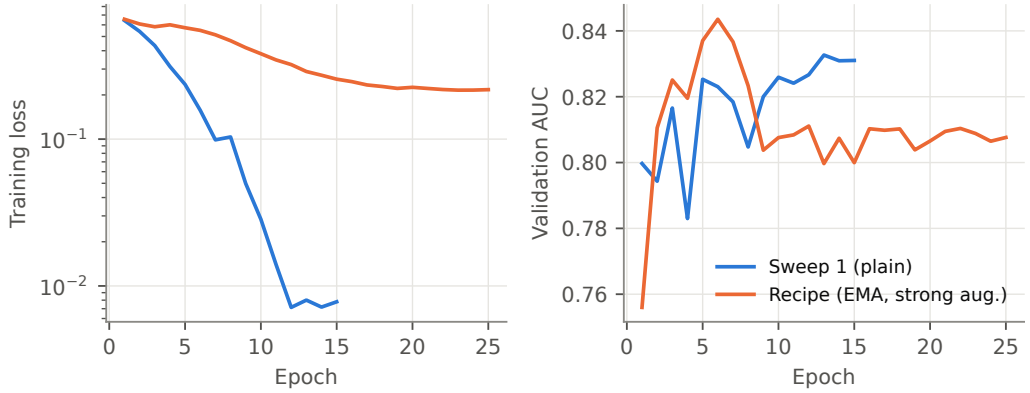



**Figure 6:** EfficientNet-B0: training loss (log scale) and validation AUC. The stronger recipe's loss stays higher because of label smoothing and augmentation, but its validation AUC peaks around epoch 6 and then declines.

# D Fusion variants and a foundation model (sweep 2)

- **Motivation.** Plain concatenation may not be the best way to combine two modalities. A large self-supervised model may transfer better than ImageNet weights.
- **Setup** (Swin-Tiny recipe, 25 epochs; Figure 7).
  - *Gated fusion* with modality dropout: $\mathbf{z} = \sigma(W[\mathbf{p}_S; \mathbf{p}_O]) \odot \mathbf{p}_S + (1 - \sigma(\cdot)) \odot \mathbf{p}_O$, with $\mathbf{p} = W'\mathbf{z}$. One modality's feature is set to zero for 20% of training samples.
  - *Cross-attention fusion* with ViT-Small: each modality's tokens attend to the other's tokens, then the class tokens are concatenated.
  - *DINOv2-Small* [18], fine-tuned, with late fusion.
- **Results** (Table 9).

| EfficientNet-B0 variant | Val AUC | Test AUC | TTA AUC |
|---|---|---|---|
| Recipe (3 seeds, mean ± s.d.) | 0.832 | 0.798 ± 0.003 | 0.803 |
| EMA off | 0.817 | 0.786 | 0.792 |
| Head LR ×1 (instead of ×5) | 0.828 | 0.807 | 0.807 |
| Flip + mild jitter only (no rotate/scale/shift) | 0.853 | 0.819 | 0.824 |
| All three reverted, no label smoothing, 15 epochs (approx. sweep-1 setup) | 0.844 | 0.811 | 0.814 |
| Sweep-1 run (original code, one seed) | 0.833 | 0.827 | 0.828 |

**Table 7:** Single-factor ablations of the stronger recipe (EfficientNet-B0, one seed each; the recipe row is the three-seed mean). The last row is the original first-sweep run.

  - Gated fusion 0.812, cross-attention 0.795, DINOv2 0.809. Plain concatenation: 0.808 ± 0.004.
  - No variant is distinguishable from the baseline.
  - Gated fusion on EfficientNet-B0: 0.810, against 0.798 for concatenation.
- **Takeaway.** Plain concatenation is enough at this data size. The final model keeps it.

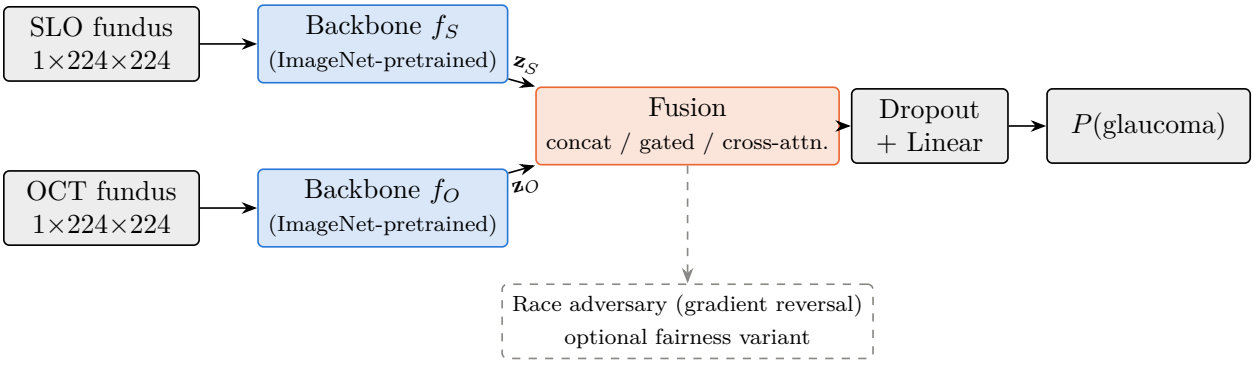



**Figure 7:** Variants evaluated in the appendices: the fusion block is replaced by gated or cross-attention fusion; a single backbone with a 2-channel input (early fusion) or with one modality only; an adversarial race head on the fused feature.

# E Fairness-aware training and per-group results (sweep 2)

- **Motivation.** The data are demographically imbalanced, and fairness-aware losses are a main theme of the dataset papers [2, 3, 1].
- **Setup** (Swin-Tiny recipe).
  - *Group reweighting*: each sample is weighted by the inverse frequency of its (race, label) cell.
  - *FIS-like loss*: weight = (sample loss / batch mean loss) × (its race group's mean loss / batch mean loss), clipped to $[0.2, 5]$ and renormalised. This is our approximation of the idea of Fair Identity Scaling [3], not an exact copy.
  - *Adversarial debiasing*: a small classifier predicts race from the fused feature through a gradient-reversal layer [28] (loss weight 0.5).
  - The adversarial variant was repeated on EfficientNet-B0.

- **Results** (Table 9, Table 8).
  - FIS-like 0.806 and adversarial 0.805: within noise of the plain recipe (0.808). Group reweighting: 0.787, lower.
  - EfficientNet-B0: adversarial 0.802, against 0.817 for the configuration it extends.
  - Gender and race groups differ by about 0.01–0.03 AUC for most models.
  - The Hispanic group (83 test images) varies most (0.74–0.81). Sampling noise could explain this.
- **Takeaway.** No fairness-specific loss gave a reliable benefit.

| $n$ (test) | Female 1449 | Male 551 | Asian 177 | Black 301 | White 1522 | Non-Hisp. 1917 | Hispanic 83 |
|---|---|---|---|---|---|---|---|
| EfficientNet-B0 (sweep 1) | 0.824 | 0.835 | 0.823 | 0.816 | 0.827 | 0.830 | 0.740 |
| Swin-T + recipe (sweep 2) | 0.813 | 0.806 | 0.832 | 0.811 | 0.806 | 0.812 | 0.782 |
| EfficientNet-B0 + drop-path (sweep 3) | 0.816 | 0.822 | 0.831 | 0.803 | 0.814 | 0.819 | 0.811 |
| 3-model ensemble, top 3 of sweep 1 | 0.817 | 0.822 | 0.827 | 0.818 | 0.811 | 0.820 | 0.779 |
| 3-model ensemble, top 3 of recipe study | 0.819 | 0.833 | 0.838 | 0.816 | 0.816 | 0.825 | 0.795 |

**Table 8:** Test AUC within demographic groups for representative models and 3-model ensembles.

# F Domain shift and tabular inputs (sweep 2)

- **Motivation.**
  - FairDomain treats image types as domains [1]. If SLO and OCT carry different information, a model trained on one should transfer poorly to the other. That would explain why fusion helps.
  - Do clinical variables add anything?
- **Setup.**
  - *Cross-domain*: one Swin-Tiny branch is trained on one modality only. It is tested on the same modality and on the other (fed through the same single-channel input).
  - *Tabular*: age (and, as an upper bound, the visual-field mean deviation MD) goes through a small MLP and is concatenated to the fused feature.
- **Results** (Table 9).
  - Trained on SLO: 0.794 on SLO, 0.677 on OCT.
  - Trained on OCT: 0.786 on OCT, 0.701 on SLO.
  - Age: 0.808, no change.
  - MD: 0.958. MD is a clinical measure closely tied to the label. This is not a fair comparison and is excluded from all other comparisons.
- **Takeaway.** The two modalities are complementary, and transfer between them is an open problem. Age adds nothing.

| Variant (Swin-Tiny backbone) | Val AUC | Test AUC | 95% CI | ES-AUC race | Worst-group | Other-domain AUC |
|---|---|---|---|---|---|---|
| + age and MD (upper bound) | 0.963 | 0.958 | [0.950, 0.965] | 0.938 | 0.941 | – |
| Gated fusion + modality dropout | 0.842 | 0.812 | [0.793, 0.832] | 0.800 | 0.803 | – |
| Swin-T + recipe | 0.834 | 0.811 | [0.790, 0.831] | 0.790 | 0.806 | – |
| Swin-T + recipe (seed 2) | 0.836 | 0.810 | [0.789, 0.830] | 0.783 | 0.793 | – |
| DINOv2-S fine-tune | 0.837 | 0.809 | [0.788, 0.829] | 0.760 | 0.786 | – |
| + age as input | 0.840 | 0.808 | [0.786, 0.828] | 0.785 | 0.802 | – |
| FIS-like loss | 0.847 | 0.806 | [0.784, 0.826] | 0.779 | 0.800 | – |
| Adversarial race debiasing | 0.850 | 0.805 | [0.782, 0.823] | 0.788 | 0.801 | – |
| Swin-T + recipe (seed 1) | 0.843 | 0.803 | [0.782, 0.823] | 0.783 | 0.797 | – |
| ViT-S cross-attention fusion | 0.844 | 0.795 | [0.773, 0.815] | 0.772 | 0.786 | – |
| Train SLO → test OCT | 0.796 | 0.794 | [0.773, 0.812] | 0.759 | 0.786 | 0.677 |
| Group-reweighted loss | 0.819 | 0.787 | [0.764, 0.808] | 0.760 | 0.780 | – |
| Train OCT → test SLO | 0.822 | 0.786 | [0.765, 0.804] | 0.758 | 0.779 | 0.701 |

**Table 9:** Sweep 2: stronger recipe on Swin-Tiny, 25 epochs. The "+ age and MD" row is an upper bound and not a fair comparison.

# G Small ensembles and more training data

- **Motivation.** Single runs overfit and are noisy. Averaging a few models, or training on more data, might help.
- **Setup.** Averages of probabilities of three models (we did not use larger ensembles):
  - Top 3 of sweep 1 by validation AUC: Swin-Tiny, ViT-Small, DenseNet121.
  - Top 3 of the recipe study by validation AUC: EfficientNet-B0 with drop-path, with the stronger recipe, and with mixup.
  - 3 seeds of EfficientNet-B0 (drop-path configuration) trained on all 3,000 training images, with no validation set; final-epoch weights.
  - Each ensemble is also reported with flip TTA.
- **Results** (Table 10).
  - Test AUC 0.816–0.823. The members average 0.792–0.806, so averaging three models gains about +0.02.
  - No 3-model ensemble beats the best single run (0.827); the differences are inside the confidence intervals.
  - Flip TTA adds 0.001–0.005.
  - More data: the three train+val models (3,000 images) score 0.791–0.794 alone, no better than models trained on 2,550 images. Five single models trained on 5-fold splits (2,400 images) score 0.789–0.808 (mean 0.800).
- **Takeaway.** Averaging three models gives a small, steady gain over their average member, but does not beat the best single run. More data did not help single models. We do not use ensembles in the final model.

# H Full result tables

- **Compute.** Single runs took about 3 minutes (ResNet18) to 18 minutes (ConvNeXt-Tiny, 15 epochs) in sweep 1, and about 11 to 22 minutes in later sweeps, on a P100. EfficientNet-B0

| Ensemble (average of probabilities) | $n$ | Members' mean AUC | Test AUC | 95% CI | TTA AUC | ES-AUC race |
|---|---|---|---|---|---|---|
| Top 3 of the recipe study by validation AUC | 3 | 0.806 | 0.823 | [0.803, 0.841] | 0.824 | 0.799 |
| Top 3 of sweep 1 by validation AUC (Swin-T, ViT-S, DenseNet121) | 3 | 0.800 | 0.819 | [0.800, 0.837] | 0.822 | 0.805 |
| 3 seeds trained on train+val (EfficientNet-B0) | 3 | 0.792 | 0.816 | [0.797, 0.834] | 0.821 | 0.764 |

**Table 10:** Three-model ensembles. "Members' mean AUC" is the average test AUC of the three members.

| Model | Type | Params (M) | Val AUC | Test AUC | TTA AUC | Acc. | Worst-group AUC |
|---|---|---|---|---|---|---|---|
| EfficientNet-B0 | CNN | 8.0 | 0.833 | 0.827 | 0.828 | 0.763 | 0.816 |
| Swin-Tiny | Transformer | 55.0 | 0.844 | 0.807 | 0.811 | 0.739 | 0.796 |
| DenseNet121 | CNN | 13.9 | 0.837 | 0.803 | 0.810 | 0.740 | 0.794 |
| CoAtNet-0 | Hybrid | 53.3 | 0.835 | 0.793 | 0.796 | 0.716 | 0.787 |
| ViT-Small | Transformer | 42.9 | 0.841 | 0.790 | 0.792 | 0.724 | 0.781 |
| MaxViT-Tiny (early fusion) | Hybrid | 28.6 | 0.820 | 0.778 | 0.788 | 0.715 | 0.768 |
| DeiT-Small (early fusion) | Transformer | 21.6 | 0.812 | 0.768 | 0.774 | 0.706 | 0.756 |
| ResNet50 | CNN | 47.0 | 0.796 | 0.768 | 0.779 | 0.703 | 0.741 |
| ConvNeXt-Tiny | CNN | 55.6 | 0.822 | 0.767 | 0.767 | 0.680 | 0.757 |
| EfficientNet-B0, OCT only | Single modality | 4.0 | 0.809 | 0.764 | 0.771 | 0.683 | 0.753 |
| EfficientNet-B0, SLO only | Single modality | 4.0 | 0.806 | 0.758 | 0.762 | 0.673 | 0.723 |
| ResNet50 (early fusion) | CNN | 23.5 | 0.746 | 0.747 | 0.759 | 0.688 | 0.745 |
| ResNet18 (scratch) | CNN | 22.3 | 0.773 | 0.717 | 0.725 | 0.666 | 0.706 |

**Table 11:** Sweep 1: 13 configurations, plain recipe, 15 epochs, validation-based epoch selection. Params = total parameters of both branches. "TTA AUC" = with horizontal-flip test-time averaging.

was the cheapest strong model.

| Configuration | Val AUC | Test AUC | 95% CI | TTA AUC | ES-AUC race | Worst-group |
|---|---|---|---|---|---|---|
| EfficientNet-B0 + recipe | 0.844 | 0.799 | [0.779, 0.820] | 0.802 | 0.774 | 0.782 |
| seed 1 | 0.818 | 0.795 | [0.774, 0.815] | 0.801 | 0.747 | 0.782 |
| seed 2 | 0.835 | 0.800 | [0.778, 0.821] | 0.805 | 0.755 | 0.777 |
| 12 epochs | 0.836 | 0.777 | [0.757, 0.798] | 0.787 | 0.760 | 0.766 |
| + mixup/CutMix | 0.839 | 0.800 | [0.780, 0.819] | 0.805 | 0.765 | 0.786 |
| + drop-path 0.2, dropout 0.5 | 0.852 | 0.817 | [0.797, 0.839] | 0.820 | 0.793 | 0.803 |
| lower LR (2e-4) | 0.833 | 0.790 | [0.767, 0.812] | 0.803 | 0.779 | 0.783 |
| gated fusion + modality dropout | 0.832 | 0.810 | [0.791, 0.830] | 0.814 | 0.801 | 0.803 |
| EfficientNet-B2 | 0.839 | 0.796 | [0.778, 0.817] | 0.806 | 0.765 | 0.783 |
| EfficientNet-B3 | 0.821 | 0.779 | [0.759, 0.799] | 0.789 | 0.738 | 0.757 |
| RegNetY-016 | 0.831 | 0.807 | [0.785, 0.828] | 0.812 | 0.773 | 0.803 |
| DenseNet121 | 0.828 | 0.781 | [0.760, 0.801] | 0.791 | 0.739 | 0.768 |
| Drop-path config + adversarial race debiasing | 0.847 | 0.802 | [0.781, 0.821] | 0.807 | 0.782 | 0.796 |

**Table 12:** Sweep 3: EfficientNet-B0 regularisation and backbone neighbours, 25 epochs unless stated. The drop-path configuration had the best validation AUC (0.852) and was used for the train+val and 5-fold runs (Appendix G).